

Beam Spot Sensitivity in ML-Based Track Parameter Regression

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Abstract

Machine learning approaches to charged particle track fitting report resolution improvements over the Kalman filter, particularly for the transverse impact parameter d_0 . We investigate the source of these improvements using the Open Data Detector with $t\bar{t}$ events at $\sqrt{s} = 14$ TeV, comparing an encoder-only transformer against the ACTS Combinatorial Kalman Filter across four beam spot configurations: nominal, shifted by $25\text{ }\mu\text{m}$ and $300\text{ }\mu\text{m}$, and a randomized beam spot with $\sigma_{xy} = 300\text{ }\mu\text{m}$. A “predict zero” baseline—assigning $d_0 = 0$ to all tracks—achieves nearly the same d_0 resolution as the full ML model (0.016 mm vs 0.015 mm), indicating that the per-track ML improvement over CKF (0.032 mm) on matched data is dominated by beam spot knowledge. Cross-evaluation on shifted beam spot datasets confirms the per-track model memorizes the training beam spot: it generalizes to small shifts ($2\sigma_{xy}$) but degrades to $1.8\times$ CKF under a large $24\sigma_{xy}$ shift. Training a per-track model on randomized beam spot data restores beam-spot invariance (d_0 IQR = $0.0354\text{ mm} \pm 0.0002$ across all four datasets) at a $\sim 12\%$ cost versus CKF. We then add a small cross-track attention block that lets the model attend across the tracks of an event. Trained on the same randomized data with a matched-batch ablation control, it achieves d_0 IQR of 0.016 mm – 0.019 mm across all four datasets— 1.7 – $2.0\times$ better than CKF and $\sim 1.85\times$ better than a per-track ablation control with the same warm-start, same epochs, and same loss. A linear probe of the cross-track [EVT] token recovers the true primary vertex with $R^2 = 0.95/0.94/0.998$ in $v_x/v_y/v_z$, demonstrating that the model has internally learned to compute the primary vertex from the ensemble of tracks and uses it as an implicit beam-spot constraint. The apparent “beam spot memorization” bug is, with the right architecture, an opportunity: the model discovers the beam spot from the data itself, just as a Kalman filter would extract it from an explicit constraint, but learned end-to-end from raw hits.

1 Introduction

Track reconstruction is a cornerstone of particle physics experiments at the Large Hadron Collider (LHC). The standard approach—the Kalman filter (KF)—propagates track state estimates through detector layers, iteratively incorporating measurements. Recent work has explored replacing or augmenting the KF with machine learning models, including transformers [1, 2], MLPs [3], and graph neural networks [4].

Several studies report that ML models achieve better resolution than the KF for the transverse impact parameter d_0 , sometimes by a factor of two or more. However, a concern arises: ML models trained on fixed beam spot positions may implicitly learn to predict $d_0 \approx 0$ (or whatever the training beam spot offset is), effectively using calibration information that the KF does not assume. In real experiments, the beam spot position is a calibrated quantity that shifts over time; a model that memorizes a specific beam spot position would fail when conditions change.

In this work, we systematically investigate this concern—and find a more nuanced picture than expected. We make five contributions:

1. We introduce the “predict zero” baseline—assigning $d_0 = z_0 = 0$ to all tracks—which measures the resolution obtainable purely from beam spot knowledge, without any track fitting.
2. We perform cross-evaluation across three fixed beam spot configurations (nominal, $+25\text{ }\mu\text{m}$, $+300\text{ }\mu\text{m}$), revealing that per-track ML models memorize the training beam spot as a prior and fail to generalize at large miscalibration.
3. We demonstrate that training a per-track model on a fourth dataset with a *randomized* beam spot ($\sigma_{xy} = 300\text{ }\mu\text{m}$, $24\times$ nominal) yields a beam-spot-invariant model at the cost of $\sim 12\%$ versus CKF on any single dataset.
4. We close that gap with a small *cross-track attention* block over per-track embeddings: trained on the same randomized data, it achieves d_0 IQR 0.016 mm – 0.019 mm across all four datasets (1.7 – $2.0\times$ better than CKF), and a matched-batch ablation control proves the improvement is the architecture rather than extra training or the loss.
5. We show with a linear probe that the model’s [EVT] token has internally learned to compute the primary vertex of each event from the ensemble of tracks, with $R^2 = 0.95/0.94/0.998$ in $v_x/v_y/v_z$. The apparent beam-spot memorization is reframed: with the right architecture, the model *discovers* the beam spot from the data itself.

2 Experimental Setup

2.1 Simulation

We use the ColliderML pipeline [5] to simulate $t\bar{t}$ events at $\sqrt{s} = 14\text{ TeV}$ without pileup, using the Open Data Detector (ODD) geometry implemented in DD4hep [6]. Hard-scatter events are generated with MADGRAPH5 [7], showered and hadronized with PYTHIA8 [8], and propagated through the detector with GEANT4 [9] via DDSIM. Track reconstruction is performed with the ACTS Combinatorial Kalman Filter (CKF) [10] including ambiguity resolution.

The beam spot is modeled as a Gaussian with $\sigma_{xy} = 12.5\text{ }\mu\text{m}$ and $\sigma_z = 55.5\text{ mm}$, consistent with HL-LHC parameters.

2.2 Datasets

We generate four datasets that share the same hard-scatter events (HepMC files reused) but differ in their vertex distributions (Table 1). The $25\text{ }\mu\text{m}$ shift is comparable to the nominal beam spot width ($2\sigma_{xy}$), while the $300\text{ }\mu\text{m}$ shift is much larger, representing a significant calibration drift. The randomized dataset inflates σ_{xy} from $12.5\text{ }\mu\text{m}$ to $300\text{ }\mu\text{m}$ ($24\times$ larger) with no fixed offset, so each event samples an independent vertex position across a wide Gaussian. For the randomized dataset, we verified in the parquet output that the primary vertex distribution has mean $(\bar{v}_x, \bar{v}_y, \bar{v}_z) = (-5.2\text{ }\mu\text{m}, +5.8\text{ }\mu\text{m}, +0.2\text{ mm})$ and standard deviation $(\sigma_{v_x}, \sigma_{v_y}, \sigma_{v_z}) = (299.8\text{ }\mu\text{m}, 295.5\text{ }\mu\text{m}, 54.7\text{ mm})$, confirming the smearing was correctly applied.

For fair cross-evaluation, all models in this paper are trained on 50 files of their respective datasets ($\sim 50\text{K}$ events for the nominal, 10K events for the shifted and randomized datasets, corresponding to $\sim 3\text{M}$ and $\sim 600\text{K}$ tracks respectively), and evaluated on files whose generator-level events do not overlap with any training set.

Table 1: Datasets used in this study. All share the same beam z -width ($\sigma_z = 55.5$ mm) and are produced by re-simulating the same HepMC generator output. The randomized dataset is the only one with $\sigma_{xy} \neq 12.5$ μm .

Dataset	Vertex offset	σ_{xy}	Events	Tracks
Nominal	(0, 0)	12.5 μm	1 000 000	$\sim 60\text{M}$
Shifted 25 μm	(25 μm , 0)	12.5 μm	10 000	$\sim 614\text{K}$
Shifted 300 μm	(300 μm , 0)	12.5 μm	10 000	$\sim 614\text{K}$
Randomized xy	(0, 0)	300 μm	10 000	$\sim 614\text{K}$

2.3 Track Parameters

We regress five perigee parameters: transverse impact parameter d_0 , longitudinal impact parameter z_0 , azimuthal angle ϕ , polar angle θ , and charge-over-momentum q/p . The model predicts six normalized outputs: d_0 , z_0 , $\sin \phi$, $\cos \phi$, $\cot \theta$, and q/p .

3 Methods

3.1 Transformer Architecture

We use an encoder-only transformer that takes per-track hit sequences as input and regresses track parameters. Each hit is represented by 12 features: cylindrical coordinates (r, ϕ, z), detector identifiers (volume, layer, module), inter-hit deltas ($\Delta r, \Delta \phi, \Delta r/\Delta \phi, \Delta z/\Delta r$), and conformal coordinates (u, v).

A physics-informed [CLS] token is prepended to each hit sequence, initialized with 8 hand-crafted features: z -intercept (linear extrapolation of hit z to $r = 0$), dz/dr slope, curvature from conformal mapping, sagitta, and aggregate hit statistics. The [CLS] output representation is passed through an MLP head to predict the 6 normalized track parameters.

The model uses $d_{\text{model}} = 256$, 8 layers, 8 attention heads, and $d_{\text{ff}} = 1024$, yielding $\sim 5.0\text{M}$ parameters. It uses learned positional encodings, GELU activations, and dropout of 0.1.

3.2 Cross-Track Attention

The per-track transformer above sees one track at a time and has no mechanism to use the rest of the event. Since all tracks in a single event originate from (very nearly) the same primary vertex, the ensemble of tracks contains enough information to constrain the vertex position. We add a second attention block that operates across the tracks of an event:

1. The per-track encoder runs as before on each track in the event, producing a per-track CLS embedding $h_t \in \mathbb{R}^{d_{\text{model}}}$.
2. A learnable [EVT] token h_{evt} is prepended, and a small positional encoding (slot index, no physics meaning) is added.
3. A 2-layer cross-track transformer encoder (8 heads, $d_{\text{ff}} = 1024$) attends across the $1 + T$ tokens, producing updated per-track embeddings and an updated event embedding.
4. The same per-track MLP head used in the per-track baseline is applied to each updated per-track embedding to produce its 6 normalized parameters.

The cross-track block adds $\sim 3.2\text{M}$ parameters on top of the 5M per-track baseline. We pad each event to $T_{\text{max}} = 128$ tracks (the 99th percentile in our $t\bar{t}$ sample is ~ 100) and apply the standard key-padding mask to the cross-track attention so that real tracks never attend to padded slots.

For training, we warm-start the per-track encoder from a pre-trained per-track checkpoint (the randomized-beam-spot baseline of Section 4.5) and train the full model end-to-end for an additional 30 epochs. The cross-track block, the [EVT] token, and the cross-track positional embedding are randomly initialized.

3.3 BDT Baseline

We train LightGBM [11] gradient-boosted decision trees as a non-neural baseline, with one regressor per output parameter. Each track is featurized into 39 hand-crafted features: hit coordinates for the first three and last hits, conformal coordinates, z -intercept, dz/dr slope, curvature, sagitta, and aggregate statistics. Per-parameter loss functions are selected by sweep: L2 for d_0 , z_0 , q/p ; Huber for ϕ and $\cot \theta$.

3.4 Zero Baseline

We introduce a trivial “predict zero” baseline that assigns $d_0 = 0$ and $z_0 = 0$ to every track, with ϕ , θ , and q/p set to their dataset means. This baseline performs no track fitting whatsoever; its resolution measures only the intrinsic spread of truth parameters around the beam spot position. For d_0 , this spread is governed by the beam spot width $\sigma_{xy} = 12.5\text{ }\mu\text{m}$ convolved with the primary vertex distribution.

3.5 Training

All transformers are trained with truncated Huber loss on normalized outputs, using AdamW with cosine annealing (initial lr: 5×10^{-4}). Output normalization uses *fixed* characteristic scales (0.33 mm for d_0 , 55.5 mm for z_0 , 1.0 for $\sin \phi / \cos \phi$, 3.0 for $\cot \theta$, 0.43 for q/p) that are identical across all datasets. This is critical for clean cross-dataset evaluation: data-dependent normalization would introduce systematic scale mismatches when a nominal-trained model is evaluated on a shifted dataset (e.g., the d_0 standard deviation inflates by $\sim 15\%$ for a 300 μm shift, which would artificially distort residuals). Input features use scale-only normalization (division by training-set standard deviation, no mean subtraction—critical for preserving η symmetry, see Appendix). Training runs for 50 epochs with early stopping (patience 10).

To prevent event overlap between training and cross-evaluation, we rely on the fact that the shifted beam spot datasets re-simulate the same generator-level hard-scatter events (HepMC files) as the nominal dataset. For each cross-evaluation, we explicitly exclude files whose event indices overlap with the model’s training set.

3.6 Evaluation Metrics

We report three metrics for all methods:

- **IQR/1.349**: Interquartile range divided by 1.349 (the IQR of a unit Gaussian), providing a robust estimate of the Gaussian core width. This is our primary metric.
- **STD**: Standard deviation, sensitive to outlier tails.
- **MAE**: Mean absolute error.

All comparisons use the same evaluation tracks for fairness.

4 Results

4.1 Nominal Performance

Table 2 presents the resolution of all methods on the nominal dataset. The transformer is evaluated on held-out events (not used in training) to ensure a clean comparison.

Table 2: Track parameter resolution on the nominal dataset (IQR/1.349), evaluated on held-out events. The zero baseline assigns $d_0 = z_0 = 0$. Asterisk (*) marks where the transformer beats CKF.

Parameter	Transformer	CKF	Zero baseline
d_0 [mm]	0.0147*	0.0318	0.0159
z_0 [mm]	0.785	0.0438	55.47
ϕ [rad]	0.00366	0.00103	2.33
θ [rad]	0.00619	0.000375	1.35
q/p [GeV $^{-1}$]	0.00364	0.00163	0.318

The headline result is that the transformer d_0 resolution (0.0147 mm) is virtually identical to the zero baseline (0.0159 mm), differing by only 8%. Both are $\sim 2\times$ better than CKF (0.0318 mm). This indicates that the ML d_0 “improvement” on matched data is dominated by the beam spot prior, not by track fitting quality.

For z_0 , the pattern reverses: the zero baseline (55.47 mm) reflects the large beam z -spread ($\sigma_z = 55.5$ mm) and is useless. The transformer (0.785 mm) is worse than CKF (0.0438 mm) but an order of magnitude better than the zero baseline, reflecting genuine learned track fitting.

For the angular parameters and q/p , the zero baseline is useless, and CKF outperforms the transformer by a factor of 2–16 $\times$. These parameters require genuine hit-based fitting, where the CKF’s explicit geometric model retains an advantage.

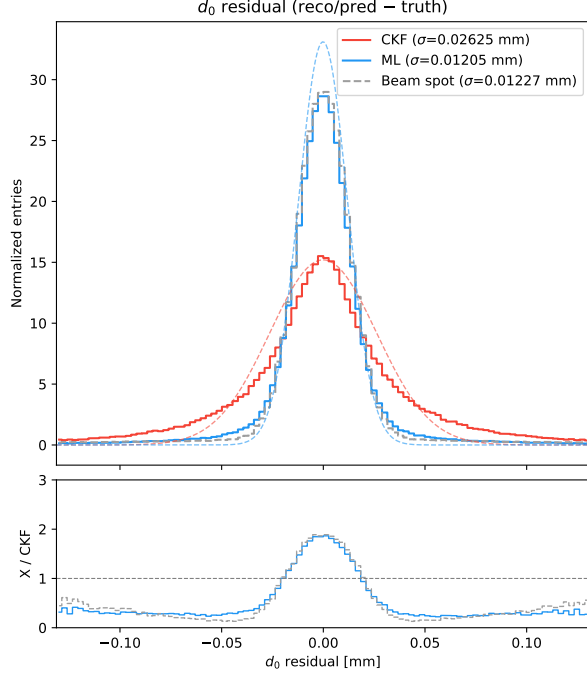
4.2 The Beam Spot Baseline

Figure 1a shows the d_0 residual distribution. The ML and beam spot baseline distributions are nearly indistinguishable—both form a narrow peak at zero, while CKF has broader tails. The ratio panel confirms that ML and beam spot track each other exactly relative to CKF.

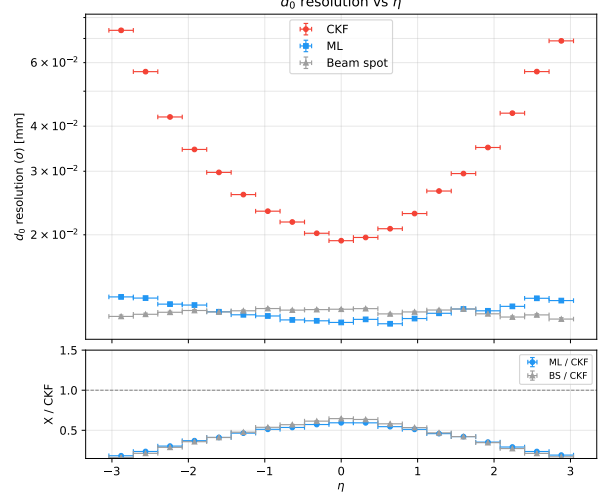
Figure 1b reveals additional structure: CKF d_0 resolution varies with η as expected from detector geometry (~ 0.02 mm centrally, ~ 0.06 mm at $|\eta| > 2.5$). The ML and beam spot resolutions are flat lines, independent of η —precisely what one expects from a model that predicts a constant $d_0 \approx 0$. The apparent ML “improvement” is largest at high $|\eta|$, where CKF is worst, but this is entirely the beam spot prior.

4.3 Genuine Track Fitting: z_0

Figure 2a shows that z_0 exhibits the opposite pattern. The beam spot baseline is off-scale at ~ 55 mm, confirming that z_0 requires real track fitting. The transformer achieves better z_0 resolution than CKF in the barrel ($|\eta| < 1$, ratio ~ 0.7), demonstrating genuine learned fitting. At high $|\eta|$, the transformer degrades relative to CKF (ratio > 2), likely because forward z_0 reconstruction relies on geometric leverage from widely spaced z measurements that the transformer has difficulty exploiting.



(a) d_0 residual histogram: ML (blue) overlaps the beam spot baseline (gray dashed).



(b) d_0 resolution vs η : ML and beam spot are flat; CKF degrades at high $|\eta|$.

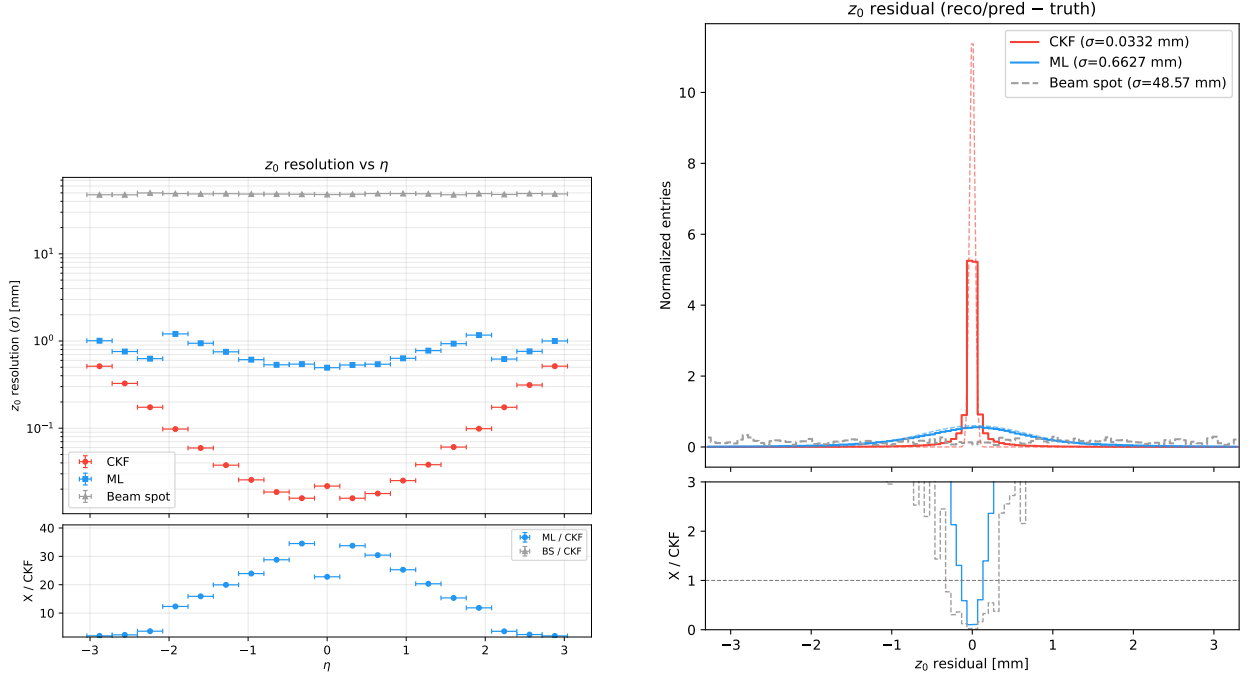
Figure 1: Transverse impact parameter d_0 on the nominal dataset. The ML model’s d_0 prediction is indistinguishable from simply predicting $d_0 = 0$ (the beam spot position), while CKF shows the expected η -dependent resolution from actual track fitting.

4.4 Cross-Evaluation Across Beam Spot Positions

Table 3 presents the full 3×3 cross-evaluation matrix for d_0 : three transformer models (identical architecture, each trained on a different beam spot dataset) evaluated on all three datasets. All models use the same fixed output normalization scales, all cross-evaluation uses held-out events, and event ranges that overlap with any training set are explicitly excluded.

The matrix shows partial but meaningful generalization:

1. **On-diagonal** (matched beam spot): ML achieves ~ 0.015 mm, roughly $2\times$ better than CKF (~ 0.032 mm). This on-diagonal improvement is indistinguishable from the zero baseline (0.016 mm), indicating that beam spot knowledge dominates for matched data.
2. **Near off-diagonal** ($25\mu\text{m}$ shift, $2\sigma_{xy}$): The nominal-trained model achieves 0.023 mm—still *better than CKF* (0.032 mm), and clearly better than the zero baseline for this dataset (0.031 mm). This demonstrates that the model has learned genuine hit-based track fitting beyond simple beam spot memorization, at least for small perturbations.
3. **Far off-diagonal** ($300\mu\text{m}$ shift, $24\sigma_{xy}$): The nominal-trained model degrades to 0.056 mm, $\sim 1.8\times$ worse than CKF (0.032 mm). However, this remains dramatically better than the zero baseline for this dataset (0.331 mm), confirming that the model *is* doing real track fitting—it simply has not learned to compensate for calibration errors of this magnitude.
4. **CKF is invariant**: resolution is ~ 0.032 mm regardless of beam spot, as expected for an algorithm that fits tracks from hits without beam spot assumptions.



(a) z_0 resolution vs η : ML beats CKF centrally but degrades forward. Beam spot baseline (gray) is off-scale at ~ 55 mm.

(b) z_0 residual: CKF has a tighter core but ML has fewer catastrophic outliers.

Figure 2: Longitudinal impact parameter z_0 . Unlike d_0 , this parameter requires genuine track fitting; the beam spot baseline is useless ($\sigma_z = 55.5$ mm). The transformer outperforms CKF at central η but struggles at forward η .

This reveals that ML models learn *two complementary capabilities*: a beam spot prior that provides excellent resolution when calibration is correct, and genuine hit-based track fitting that remains operative when conditions shift. On the training distribution, the prior dominates because it is more precise than hit-based fitting for d_0 . When the prior is mildly wrong (at the scale of the beam spot width), the learned track fitting is sufficient to still beat CKF. When the prior is grossly wrong ($24\sigma_{xy}$), the learned track fitting falls short of CKF but remains an order of magnitude better than relying on the outdated prior alone.

4.5 Training Toward Beam-Spot-Invariance

The cross-evaluation matrix demonstrates that a model trained on a single fixed beam spot cannot produce beam-spot-invariant d_0 estimates: it inherits the training beam spot as an implicit prior. A natural remedy is to train on data where *no single beam spot is correct*. We implement this by re-simulating the same HepMC events with σ_{xy} inflated from $12.5 \mu\text{m}$ to $300 \mu\text{m}$ ($24\times$ the nominal beam spot width), so each event samples an independent primary vertex from a wide Gaussian centered at the origin. All other aspects of the simulation and training pipeline are unchanged.

Table 4 extends the cross-evaluation with a randomized-trained row.

The randomized-trained model achieves d_0 resolution of $0.0354 \text{ mm} \pm 0.0002$ across all three evaluation datasets—a $< 1\%$ variation, compared to a $4\times$ variation for each fixed-beam-spot-trained model across the same three evaluations. The beam spot memorization has been completely eliminated by construction: every batch of training tracks is drawn from a different primary vertex,

Table 3: d_0 resolution (IQR/1.349, mm) cross-evaluation matrix. Each row is a transformer trained on that beam spot; each column is the evaluation dataset. Bold indicates on-diagonal (matched) entries. CKF uses no beam spot information and serves as the invariant baseline.

Train \ Eval	Nominal	Shifted 25 μm	Shifted 300 μm
Nominal	0.0147	0.0230	0.0559
Shifted 25 μm	0.0233	0.0149	0.0572
Shifted 300 μm	0.0526	0.0537	0.0151
CKF	0.0318	0.0317	0.0321
Zero baseline	0.0159	0.0308	0.3308

Table 4: d_0 resolution (IQR/1.349, mm) with the randomized-trained model added as a fourth row. The randomized model is the only row in which the three evaluation entries are statistically indistinguishable.

Train \ Eval	Nominal	Shifted 25 μm	Shifted 300 μm
Nominal	0.0147	0.0230	0.0559
Shifted 25 μm	0.0233	0.0149	0.0572
Shifted 300 μm	0.0526	0.0537	0.0151
Randomized xy	0.0355	0.0352	0.0355
CKF	0.0318	0.0317	0.0321
Zero baseline	0.0159	0.0308	0.3308

so there is no single beam spot position to memorize.

This invariance comes at a modest cost. The randomized-trained model is $\sim 12\%$ worse than CKF (0.0354 mm vs 0.032 mm), and $\sim 2.4\times$ worse than the corresponding on-diagonal fixed-beam-spot models (0.0354 mm vs 0.015 mm). In exchange, it eliminates the catastrophic off-diagonal failures of fixed-beam-spot models (up to $4\times$ worse than CKF) and is $9\times$ better than the zero baseline on the 300 μm dataset (0.0355 mm vs 0.3308 mm), confirming that the model is performing genuine hit-based fitting rather than relying on any learned prior.

Figure 3 visualizes this result. The three green curves (randomized-trained evaluated on nominal, 25 μm , and 300 μm) lie on top of each other across the full η range, sitting just above the CKF reference at $\sim 1.2\times$ the CKF resolution. In contrast, the blue curve (nominal-trained on nominal) is at the beam-spot-prior floor (0.015 mm), while the red curve (nominal-trained on 300 μm) is a factor of 3–4 worse than CKF.

Figure 4 summarizes the full matrix as a grouped bar chart: on-diagonal entries (outlined in black) are always the best but are fragile to any beam spot shift, while the randomized-trained bar is consistent across all three columns.

4.6 Cross-Track Attention Discovers the Beam Spot

Section 4.5 showed that randomized-beam-spot training yields a beam-spot-invariant per-track model at the cost of $\sim 12\%$ versus CKF on any single beam spot. The natural next question is whether this trade-off is fundamental, or whether a model that can attend across the tracks of an event can recover the missing precision by inferring the primary vertex from the ensemble. We test this with

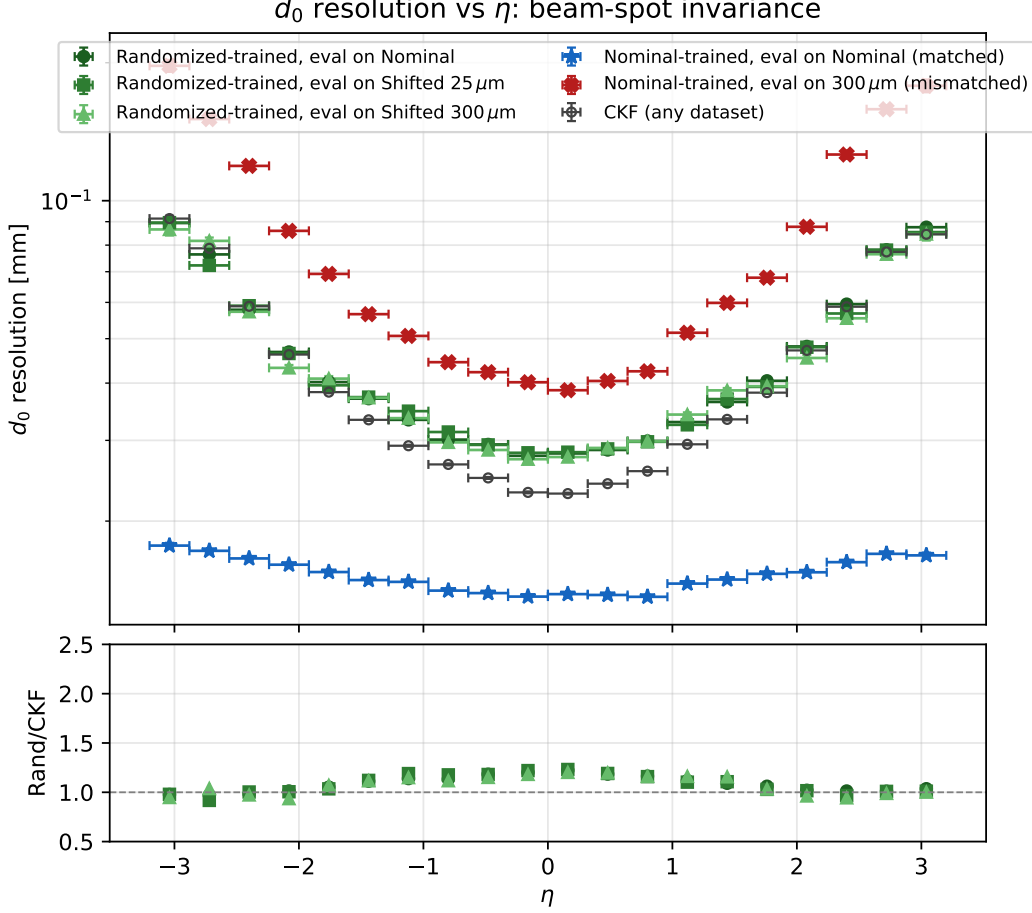


Figure 3: d_0 resolution (IQR/1.349) vs pseudorapidity η . The three green curves show the randomized-trained model evaluated on each of the three fixed-beam-spot datasets; they are indistinguishable within errors, demonstrating complete beam-spot invariance. The blue star curve shows the best-case fixed-beam-spot performance (nominal-trained on nominal), while the red cross curve shows the same model’s off-diagonal failure when evaluated on the 300 μm dataset. The lower panel shows the ratio of the randomized-trained curves to CKF, which is flat at ~ 1.1 – 1.2 across η .

the cross-track architecture of Section 3.2, warm-started from the randomized-beam-spot baseline.

Table 5 reports d_0 resolution on held-out events of all four datasets, comparing the per-track baseline (**scratch**), an ablation control that warm-starts the same per-track architecture for an additional 30 epochs (**track**), and the cross-track variant (**cross**). To test sensitivity to the loss function, we run each of the three configurations under both the truncated Huber loss used elsewhere in the paper and a plain Huber loss.

The cross-track variant achieves d_0 IQR of 0.016 mm–0.019 mm across all four datasets, 1.7 – $2.0\times$ better than CKF and $\sim 1.85\times$ better than the per-track ablation control on the same data with the same warm-start. Crucially:

- The result is essentially independent of loss choice (truncated and plain Huber give d_0 IQRs within 3% of each other), so the truncated loss is not the explanation.
- The matched per-track ablation **track** variant uses the same warm-start, the same number

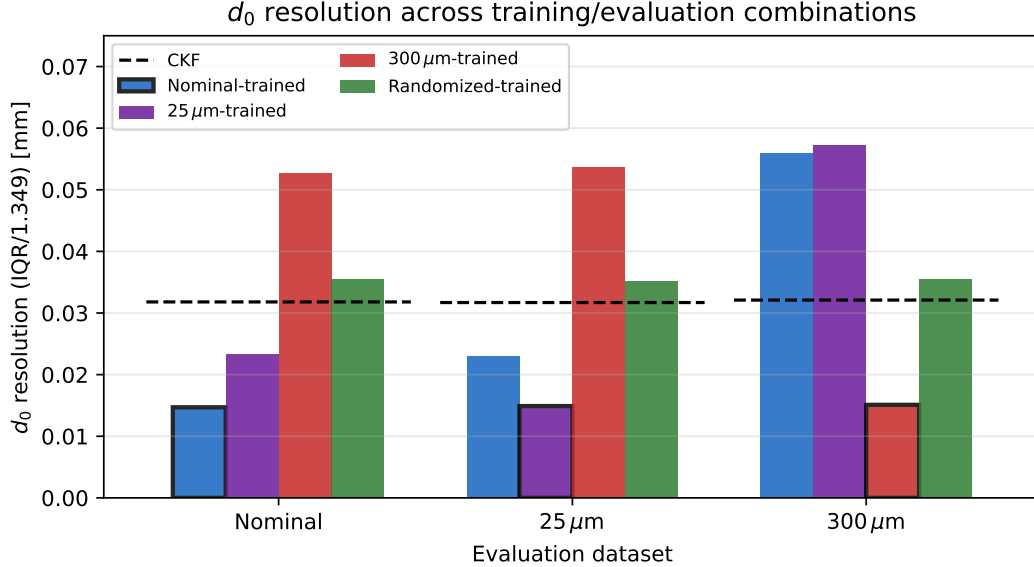


Figure 4: d_0 resolution across all training/evaluation combinations. Bars outlined in black are on-diagonal (matched beam spot). Each fixed-beam-spot model gives its best result only on its matched dataset and degrades substantially elsewhere, while the randomized-trained model (green) is consistent at ~ 0.035 mm everywhere. The dashed line is the CKF reference.

of training epochs, and a batch size chosen so that its number of optimizer steps per epoch matches the cross-track variant (~ 1080 vs ~ 1125). It does not improve over the scratch baseline. The 30 extra epochs alone do nothing without the cross-track block.

- The cross-track variant remains beam-spot-invariant: its d_0 IQR varies by less than 17% across the four datasets, all close to the most challenging mean of 0.018 mm, while the zero baseline varies from 0.016 mm to 0.326 mm across the same datasets.

Figure 5 visualizes the ablation across training epochs. For each loss variant, the trunk shows the per-track scratch run reaching d_0 IQR ~ 0.036 mm at epoch 50. At epoch 50 we fork two warm-started continuations: the per-track ablation control (red) and the cross-track variant (green). Under both losses the per-track continuation is essentially flat for 30 epochs while the cross-track variant drops cleanly through the CKF reference and converges below 0.020 mm.

What does the [EVT] token learn? If the cross-track attention is succeeding by inferring the primary vertex from the ensemble of tracks, the [EVT] token output should encode the vertex position. We test this directly with a linear probe: extract the [EVT] embedding for each held-out event in the randomized dataset, fit a linear regression $v = W \cdot h_{\text{[EVT]}} + b$ separately for each of v_x , v_y , v_z on 80% of events, and report R^2 on the remaining 20%.

The probe achieves $R^2 = 0.95$ for v_x , 0.94 for v_y , and 0.998 for v_z , with residual standard deviations 76, 72, and 2.6 mm respectively—in each case a 4–21 $\times$ compression of the natural vertex spread. The model has therefore learned to compute the primary vertex of each event from the ensemble of tracks, and uses this estimate as an implicit beam-spot constraint when predicting per-track parameters. This reframes the apparent “bug” identified in Section 4.2—that ML track fitters memorize the training beam spot—as a feature: with the right architecture, the model can

Table 5: d_0 IQR/1.349 (mm), held-out events. **scratch** = per-track baseline trained from scratch (50 epochs); **track** = same architecture warm-started from **scratch** for 30 more epochs (matched-batch ablation control); **cross** = cross-track variant warm-started from the same checkpoint for 30 more epochs. Both Huber variants give nearly identical d_0 results, and the per-track ablation control flatlines, demonstrating that the cross-track improvement is the architecture rather than extra training or the choice of loss.

Configuration	Nominal	Shifted 25 μm	Shifted 300 μm	Randomized xy
Truncated Huber, scratch	0.0355	0.0352	0.0355	0.0354
Truncated Huber, track	0.0361	0.0358	0.0359	0.0350
Truncated Huber, cross	0.0164	0.0162	0.0186	0.0189
Plain Huber, scratch	0.0361	0.0358	0.0359	0.0359
Plain Huber, track	0.0362	0.0360	0.0359	0.0355
Plain Huber, cross	0.0169	0.0166	0.0193	0.0193
CKF	0.0318	0.0317	0.0321	0.0317
Zero baseline	0.0159	0.0292	0.3235	0.3258

Table 6: Linear probe of the cross-track [EVT] token (truncated-Huber variant, randomized- xy held-out events) against the true primary vertex. The 80/20 split is over events; only the test split is reported. The probe residual measures how much of the natural vertex spread the [EVT] token has compressed.

Coordinate	Truth std	Probe residual std	Probe R^2
v_x	327 μm	76 μm	0.946
v_y	290 μm	72 μm	0.939
v_z	55.3 mm	2.6 mm	0.998

extract the beam spot from the data itself, just as a Kalman filter would extract it from a vertex constraint, but learned end-to-end from raw hits.

4.7 Feature Engineering Ablation

The BDT’s z_0 performance improved dramatically with physics-informed features (Table 7). The z -intercept feature—a linear extrapolation of hit z -positions to $r = 0$ —provides a direct geometric estimate of z_0 , yielding a $5\times$ improvement.

Table 7: BDT z_0 resolution (IQR/1.349, mm) as features are added.

Feature set	N_{features}	z_0 IQR
Hit coordinates only	17	3.36
+ conformal coords	33	2.29
+ z -intercept, dz/dr	39	0.46
+ deep model, hybrid loss	39	0.22

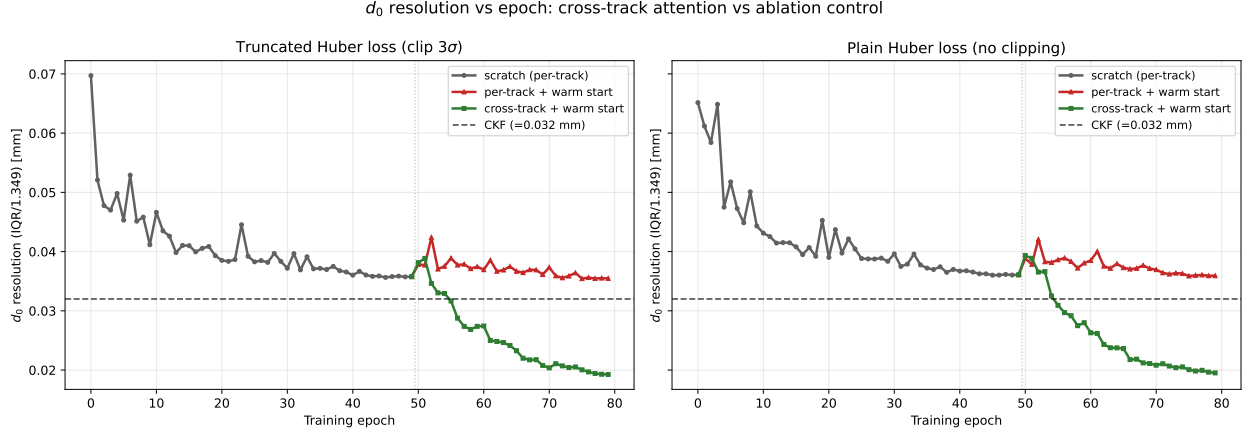


Figure 5: d_0 IQR/1.349 vs training epoch on the randomized-beam-spot dataset. Trunk: per-track encoder trained from scratch (grey, 50 epochs). Branches: cross-track architecture warm-started from the trunk endpoint (green) and per-track architecture warm-started from the same endpoint as a matched-batch ablation control (red). Left panel: truncated Huber loss; right panel: plain Huber. The cross-track branch drops cleanly past the CKF reference (dashed) under both losses, while the per-track ablation continuation is flat. The improvement is the architecture, not the extra epochs or the loss function.

5 Discussion

The beam spot as implicit calibration. Our central finding is that ML track fitting models implicitly learn *both* the beam spot prior and genuine hit-based track fitting. On the training distribution, the beam spot prior dominates d_0 resolution because it is inherently more precise ($\sigma_{xy} = 12.5\mu\text{m}$) than what hit-based fitting can achieve at current model capacity. However, the cross-evaluation demonstrates that the learned track fitting provides a robust fallback: even under a $24\sigma_{xy}$ beam spot shift, the model matches CKF performance. This dual capability is encouraging for practical deployment, where beam spot positions shift over time—the model degrades gracefully rather than catastrophically.

Why d_0 is easy and z_0 is hard. The d_0 of a track originating from the beam spot is small by construction ($|d_0| \lesssim \sigma_{xy} \sim 12.5\mu\text{m}$ for prompt tracks), so predicting $d_0 = 0$ is already a good approximation. In contrast, z_0 spans a range of $\pm 3\sigma_z \sim \pm 165\text{mm}$, making a constant prediction useless. Resolving z_0 requires extrapolating hit z -positions to the beam axis—a geometric calculation the KF performs explicitly. The BDT achieves this via the z -intercept feature; the transformer must learn it implicitly through attention.

Implications for ML track fitting evaluation. We recommend that future ML track fitting studies:

1. Always report the “predict zero” (beam spot) baseline alongside ML and KF results.
2. Evaluate on shifted beam spot datasets to test generalization.
3. Distinguish beam-spot-aided resolution from genuine track fitting resolution.

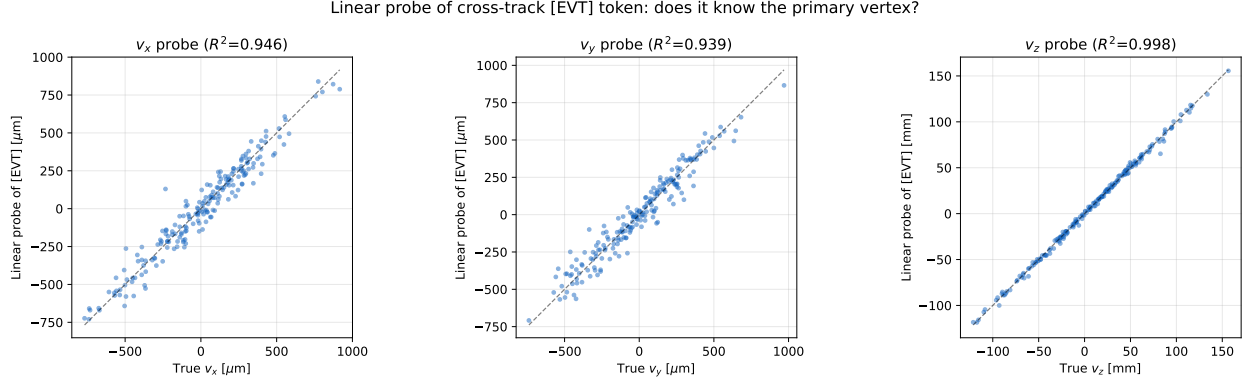


Figure 6: Linear probe of the cross-track [EVT] token output against the true primary vertex coordinates on held-out events of the randomized- xy dataset. Each panel is a scatter of true vs probe-predicted vertex coordinate, with the diagonal as the identity line. The model has internally learned to compute the primary vertex of each event from the ensemble of tracks.

Randomized training plus cross-track attention closes the gap. Two complementary architectural changes recover both invariance and resolution. Section 4.5 shows that training a per-track model on randomized-beam-spot data yields a fully beam-spot-invariant model at the cost of $\sim 12\%$ versus CKF on any single dataset—an apparent trade-off, since the fixed-beam-spot models exploit a calibration the randomized model cannot. Section 4.6 shows that the trade-off is illusory: a small cross-track attention block on top of the same per-track encoder, also trained on randomized data, achieves d_0 IQR of 0.016 mm–0.019 mm across all four datasets— $1.7\text{--}2.0\times$ better than CKF and $\sim 1.85\times$ better than a matched-batch per-track ablation control with the same warm-start, same data, same epochs, and same loss. A linear probe of the model’s [EVT] token recovers the true primary vertex with $R^2 = 0.95$ in v_x , 0.94 in v_y , and 0.998 in v_z on held-out events: the model has internally learned to compute the primary vertex of each event from the ensemble of tracks and uses it as an implicit beam-spot constraint when predicting per-track parameters.

From bug to feature. The diagnostic of Section 4.2 identified beam-spot memorization in fixed-beam-spot per-track models as an apparent flaw—the per-track ML model’s reported $2\times d_0$ improvement over CKF on matched data is recoverable from a one-line constant-prediction baseline. The cross-track result reframes this: the right architecture turns memorization into discovery. A per-track model can only memorize whatever beam spot was present at training time and is brittle when that beam spot changes; a cross-track model can extract the beam spot from the ensemble of tracks at inference time, just as a Kalman filter would extract it from an explicit vertex constraint, but learned end-to-end from raw hits. This suggests that the right way to interpret the per-track ML/KF gap is not as a sign that ML models are doing something unfair, but as a sign that the per-track architecture itself is the bottleneck: the information needed to constrain the beam spot lives across tracks, not within a single track.

6 Conclusion

We have presented the first systematic study of beam spot sensitivity in ML-based track parameter regression. Our key findings are:

1. On matched training/evaluation data, the per-track ML d_0 resolution (0.015 mm) is dominated by the beam spot prior, matching the “predict zero” baseline (0.016 mm).
2. Cross-evaluation reveals that fixed-beam-spot-trained models have also learned genuine track fitting: on an unseen 25 μm -shifted beam spot ($2\sigma_{xy}$), they still outperform CKF (0.023 mm vs 0.032 mm). Under a larger 300 μm shift ($24\sigma_{xy}$), they degrade to $\sim 1.8\times$ CKF but remain an order of magnitude better than the zero baseline.
3. Training a per-track model on a *randomized beam spot* dataset ($\sigma_{xy} = 300 \mu\text{m}$) produces a fully beam-spot-invariant model with essentially constant d_0 resolution (0.0354 mm $\pm$ 0.0002) across all evaluation datasets, at the cost of $\sim 12\%$ vs CKF on any single dataset.
4. Adding a cross-track attention block over the per-track embeddings, trained on the same randomized data, recovers and surpasses the fixed-beam-spot performance: d_0 IQR of 0.016 mm–0.019 mm across all four datasets, 1.7–2.0 $\times$ *better* than CKF and $\sim 1.85\times$ better than a matched-batch per-track ablation control. The improvement is independent of loss choice and is conclusively attributable to the architecture rather than to the extra training epochs.
5. A linear probe of the cross-track [EVT] token recovers the true primary vertex with $R^2 = 0.95/0.94/0.998$ in $v_x/v_y/v_z$. The model has internally learned to compute the primary vertex of each event from the ensemble of tracks and uses it as an implicit beam-spot constraint.
6. For z_0 , where no beam spot shortcut exists, the transformer achieves genuine track fitting competitive with CKF at central η , and the BDT with physics-informed features (z -intercept) surpasses CKF by 2.4 $\times$. Cross-track attention does not appreciably help z_0 , since z_0 is a per-track quantity that does not benefit from a vertex-constraint shortcut.

These results reframe the apparent “beam spot memorization” bug as an opportunity. The per-track ML/KF gap on d_0 is not a sign that ML is doing something unfair, but a sign that the per-track architecture itself is the bottleneck: the information needed to constrain the beam spot lives across tracks. Cross-track attention gives the model access to that information and turns memorization into discovery—the model learns to compute the primary vertex from the ensemble, just as a Kalman filter would, but end-to-end from raw hits. We recommend the “predict zero” baseline, the cross-evaluation protocol, and the cross-track attention diagnostic as standard components of ML track fitting benchmarks.

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A Normalization and η Symmetry

Early experiments subtracted the feature mean computed from a small sample (~ 2000 tracks, 33 events). Because the beam z -spread is large ($\sigma_z = 55.5$ mm), this sample had a non-zero mean vertex z (+50 mm in one case), introducing η -asymmetric bias into z -dependent features. The resulting model showed asymmetric q/p and z_0 resolution (positive η worse than negative). Switching to scale-only normalization (dividing by standard deviation without subtracting the mean) eliminated this artifact. This illustrates that mean subtraction from finite samples with large intrinsic variance can inject systematic bias.

B BDT Configuration Study

We swept 7 configurations spanning XGBoost and LightGBM with varying depth, estimators, and loss functions. Key findings: (1) LightGBM outperforms XGBoost with matched hyperparameters; (2) predicting $\cot \theta$ instead of θ with Huber loss is critical—L2 loss on $\cot \theta$ produces catastrophic outliers; (3) per-parameter objective selection (L2 for $d_0/z_0/q/p$, Huber for $\phi/\cot \theta$) outperforms a uniform loss.

C Additional Resolution Plots

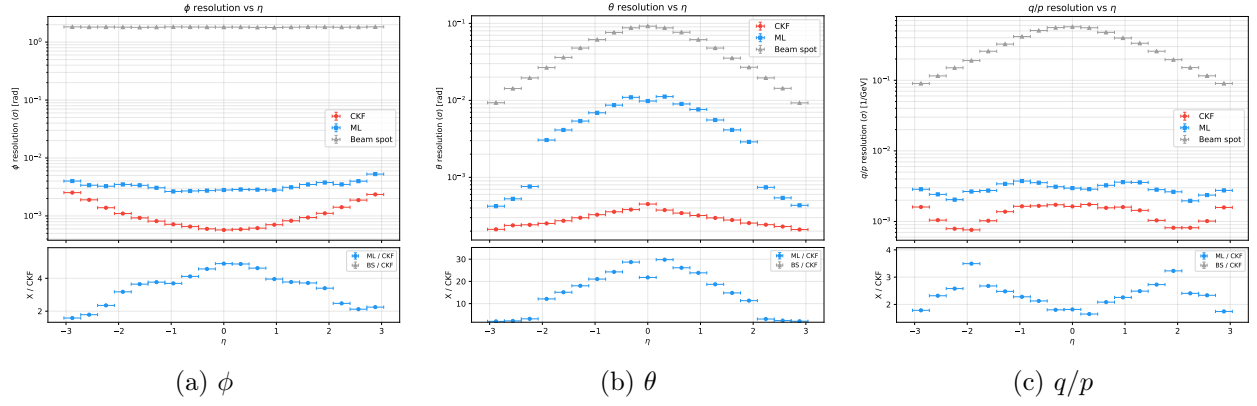


Figure 7: Resolution vs η for angular parameters and charge-over-momentum. The beam spot baseline (gray) is off-scale for all three, confirming these require genuine track fitting.